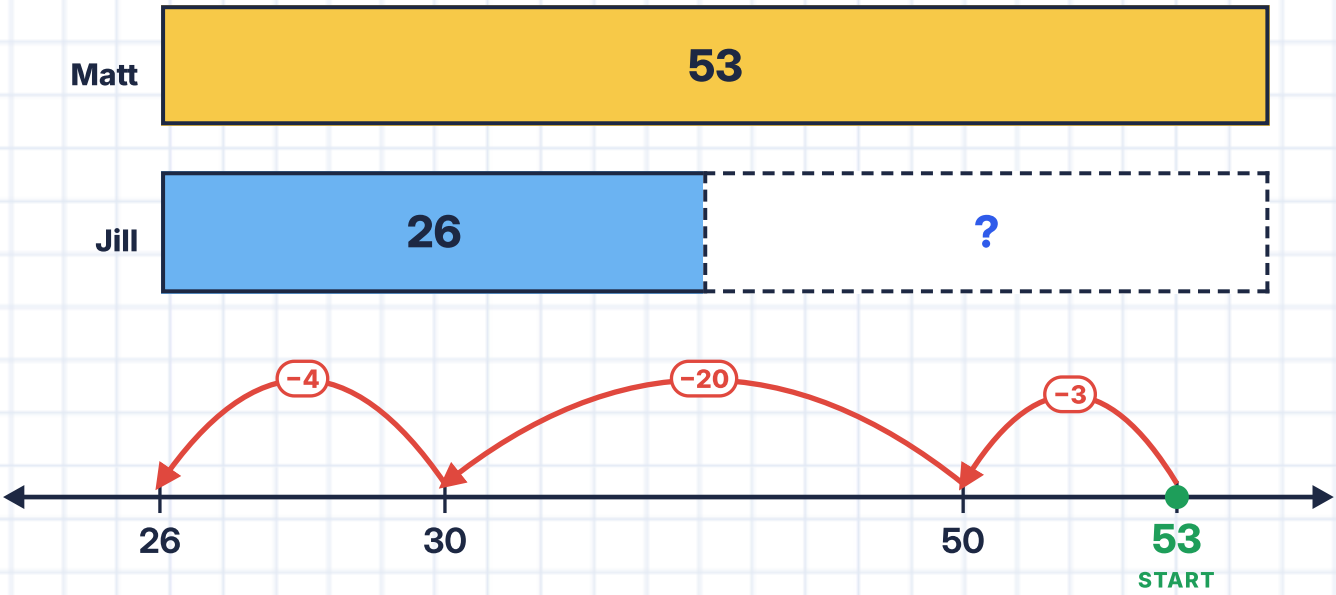
START HERE

Finding the Difference

How to find **how many more**, **how much farther** or **the difference** with a tape diagram and a number line, using **benchmark numbers**. This follows the way your Eureka Math² book does it (Grade 2, Module 1, Lessons 17 and 18).



3 + 20 + 4 = 27 miles more

My name: _____ Class: _____

Two kinds of questions

Before you draw, find out which kind of question it is. They are solved in different ways!

KIND 1 · HOW MANY MORE? HOW MUCH FARTHER? WHAT IS THE DIFFERENCE?

You already know **both numbers**. You hop from one number to the other.

The answer = all the hops added together.

Example: Matt rides 53 miles. Jill rides 26 miles. How many more miles does Matt ride? → hop from 53 to 26 → hops 3 + 20 + 4 = **27**.

KIND 2 · TAKE AWAY A NUMBER

You know the start and how much to take away. You hop back that much.

The answer = the number where you land.

Example: 181 − 14 → start at 181, hop back 14 in parts → land on **167**.

CLUE WORDS FOR KIND 1

how many more · how many fewer · how much farther · how much longer · how much taller · difference

YOUR TURN: WRITE 1 OR 2

a) Riya has 45 beads. Sam has 28. How many more beads does Riya have?

_____

b) There are 64 birds. 15 fly away. How many are left? _____

c) A red ribbon is 72 cm. A blue ribbon is 39 cm. How much longer is the red ribbon? _____

STEP 2

Benchmark numbers

Benchmark numbers are friendly numbers that end in **0**: 10, 20, 30, 40, 50 ... 180, 190. It is easy to jump to them and from them.



26 is between 20 and 30. 53 is between 50 and 60.

FIND THE NEXT TEN

Number	Ten just below	Ten just above
53	50 (3 back)	60 (7 forward)
26	20 (6 back)	30 (4 forward)
57	50 (7 back)	60 (3 forward)
181	180 (1 back)	190 (9 forward)

REMEMBER

To get to a benchmark, look at the **ones digit**. 53 → take away the 3 → 50. 26 → add 4 to make 30.

YOUR TURN

Number	Ten just below	Ten just above
41	_____	_____
18	_____	_____
67	_____	_____
35	_____	_____

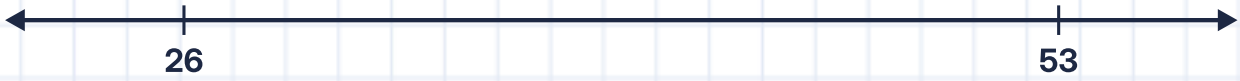
How to draw the number line

On a number line, numbers always go from **small on the LEFT** to **big on the RIGHT**, just like a ruler.

1. **Draw a straight line with an arrow at each end.**

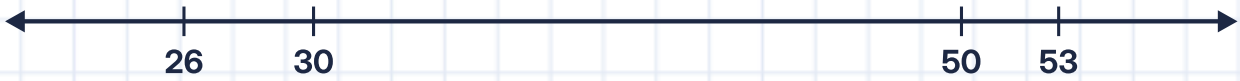
2. **Write the BIGGER number near the RIGHT end. Write the SMALLER number near the LEFT end.**

   Matt 53 and Jill 26 → 53 on the right, 26 on the left.



3. **Write the benchmarks in between.**

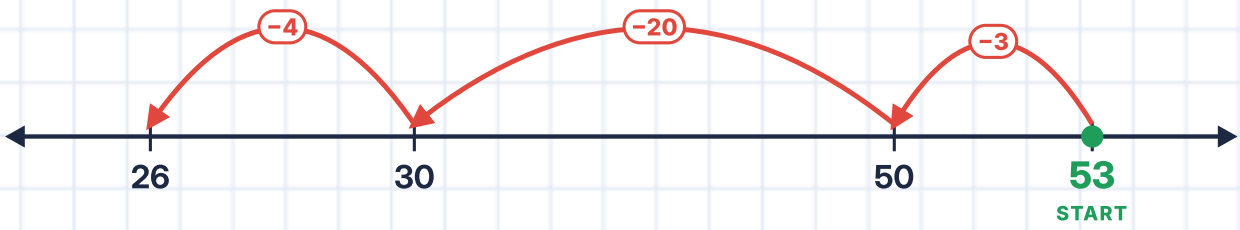
   The ten just below the big number (53 → **50**) goes a little to the left of 53. The ten just above the small number (26 → **30**) goes a little to the right of 26.



Check: read left to right, the numbers get bigger: 26, 30, 50, 53 ✔

4. **Draw the hops with arrows.**

   Hop back (Lesson 18): start at 53, arrows point LEFT. Count on (Lesson 17): start at 26, arrows point RIGHT.



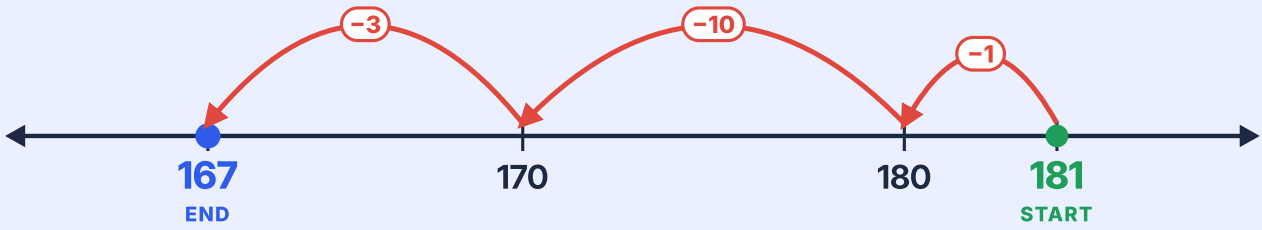
Which number goes on which side?

Small on the **left**, big on the **right**. Always.

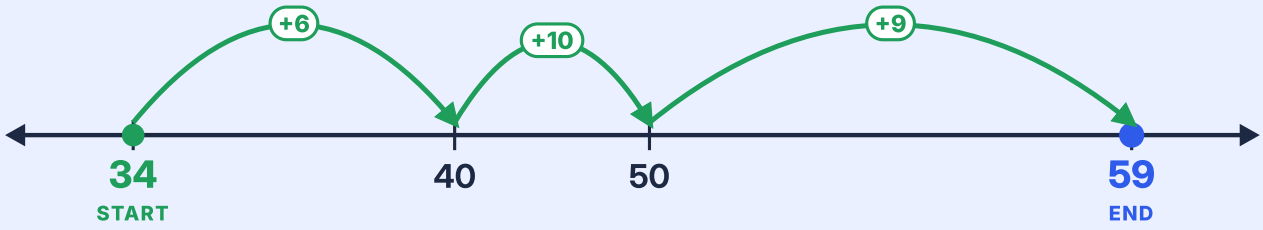
QUICK RULES

Question	Right side	Left side	Arrows
How many more? 53 and 26	53 (bigger)	26 (smaller)	← hop back, or → count on
Take away: 181 − 14	181 (start)	the answer, 167	← left
Add: 34 + 25	the answer, 59	34 (start)	→ right

TAKE AWAY: 181 − 14 (START ON THE RIGHT, GO LEFT)



ADD: 34 + 25 (START ON THE LEFT, GO RIGHT)



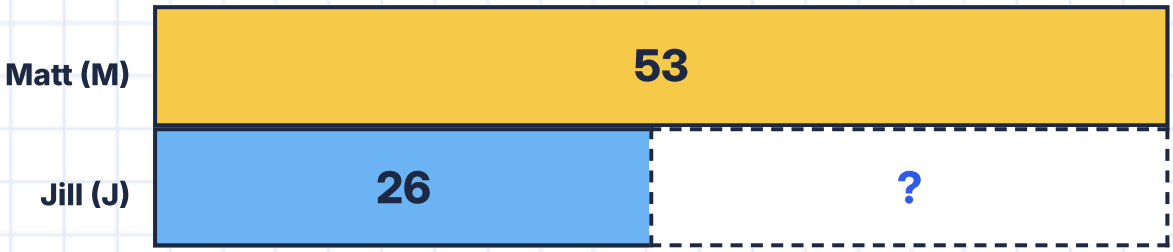
YOUR TURN

Jack 41 and Beth 18. Which number goes on the right? _____ On the left? _____ Which benchmarks go in between? _____ and _____

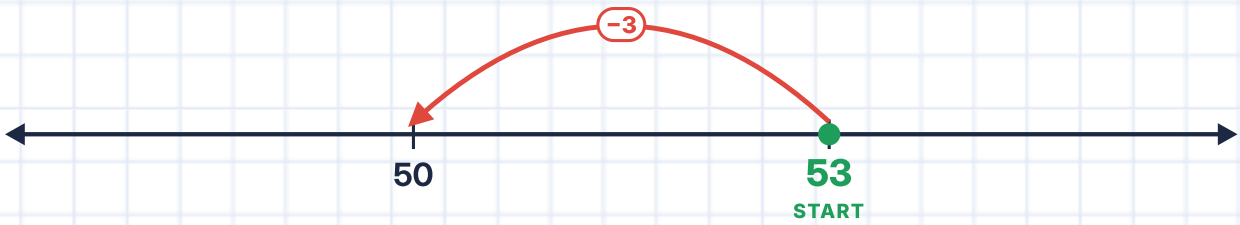
Kind 1: hop back from the bigger number

Matt rides 53 miles on a train. Jill rides 26 miles on a bus. How many more miles does Matt ride than Jill?

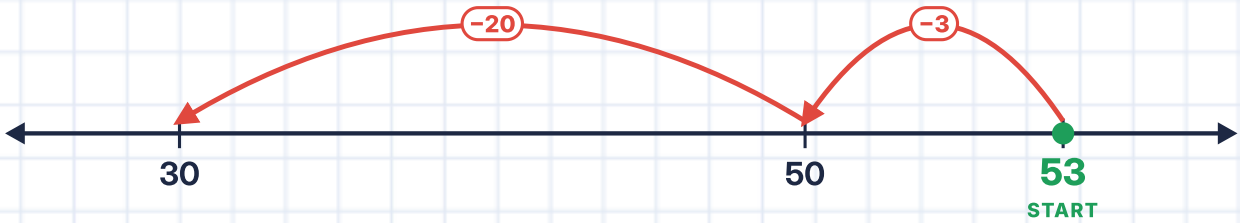
1 DRAW a tape for each person. Jill's tape is shorter. The dashed part with ? is the difference.



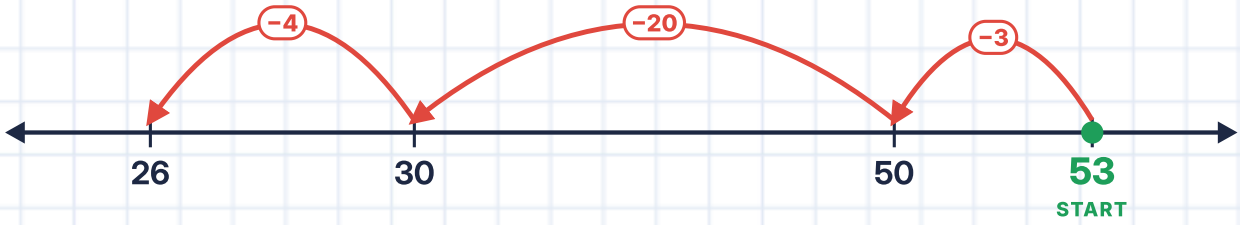
2 Draw a number line. START at the bigger number, 53. Hop back to a benchmark: 53 → 50. That hop is 3.



3 Hop back by tens to the ten next to 26: 50 → 30. That hop is 20.



4 Hop back to 26: 30 → 26. That hop is 4.

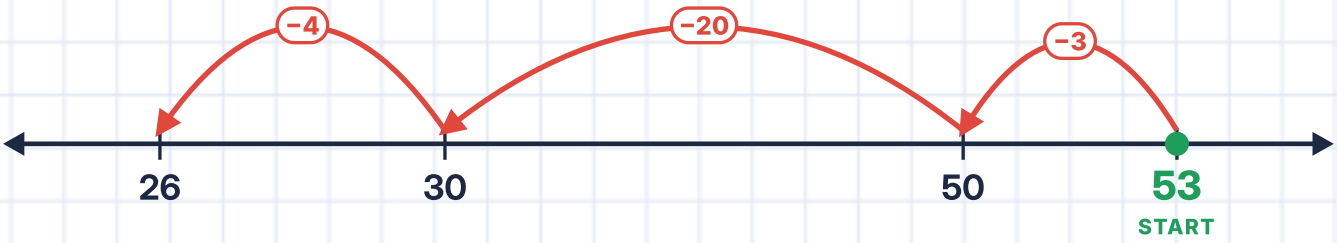


5 ADD all the hops. That is the difference.

3 + 20 + 4 = 27

STEP 3 · WRITE

Writing the answer



I hopped back **3 + 20 + 4 = 27** altogether.

WRITE THE EQUATION

Matt's 53 miles, minus the extra 27, leaves Jill's 26 miles:

53 − 27 = 26 or **26 + 27 = 53**

Answer sentence: **Matt rides 27 more miles than Jill.**

THE BIG MISTAKE

Hoppy **lands on 26**, but 26 is **Jill's number**, not the answer!

In a "how many more" question, the answer is **how far you hopped**: 27.

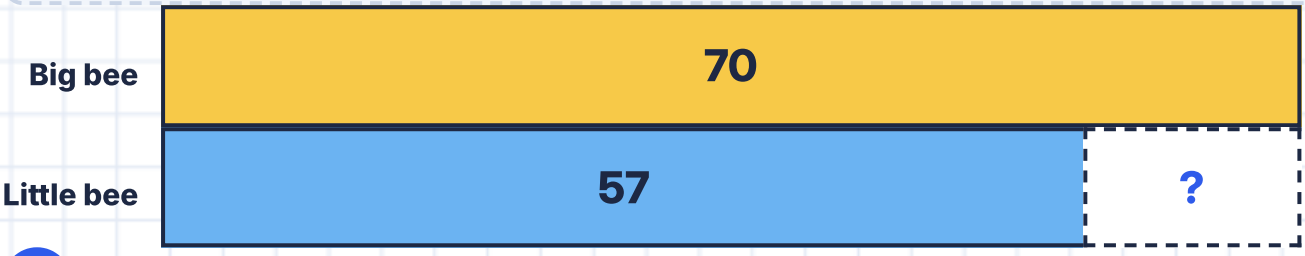
REMEMBER

How many more? → add up the hops.

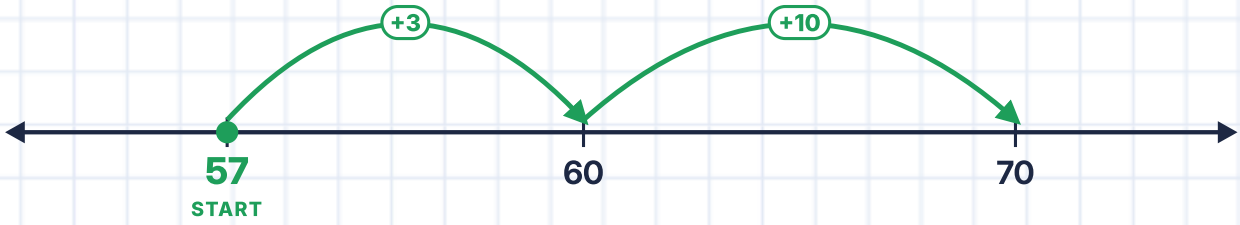
Kind 1: count on from the smaller number

You can also **start at the smaller number** and hop **forward** to the bigger one.

How much farther does the big bee fly than the little bee? Little bee: 57 cm. Big bee: 70 cm.



1. START at 57. Hop forward to a benchmark: 57 → 60. That hop is 3.
2. Hop forward by tens: 60 → 70. That hop is 10. We are at 70, so we stop.



3. ADD the hops.

3 + 10 = 13 **57 + 13 = 70**

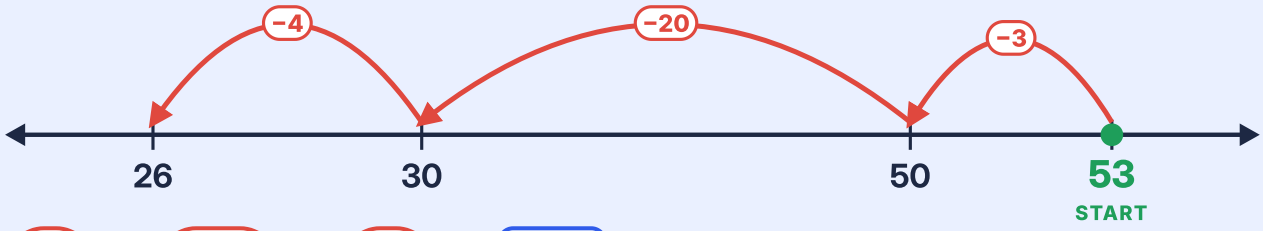
The big bee flies 13 cm farther than the little bee.

STEP 5

Back or on? Same answer!

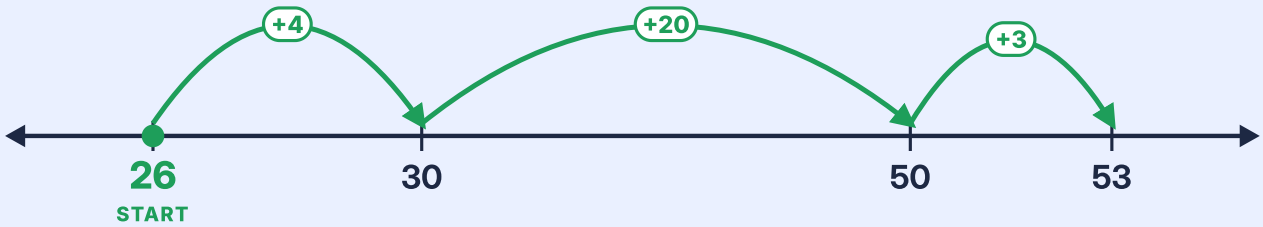
For a "how many more" question you may hop **back from the bigger** number or **on from the smaller** number. The answer is the same.

HOP BACK FROM 53 TO 26



3 + 20 + 4 = 27

HOP ON FROM 26 TO 53



4 + 20 + 3 = 27

REMEMBER

Both ways: **27**. Use the way your teacher asks for. The book uses **hop back** in Lesson 18 and **count on** in Lesson 17.

Kind 2: take away with a benchmark

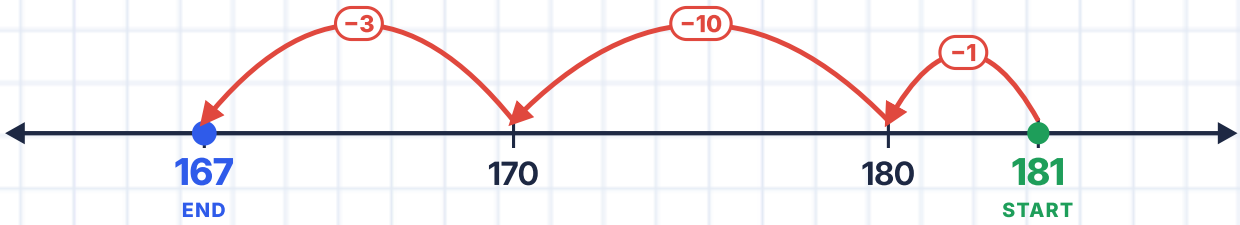
181 − 14 (The blue butterfly flies 181 cm. It flies 14 cm farther than the green butterfly. How far does the green butterfly fly?)

1. **START at the first number, 181.**

2. **Break 14 so you reach a benchmark first. 181 → 180 needs 1. So 14 = 1 + 10 + 3.**

   1 + 10 + 3 = 14

3. **Hop back 1 → 180. Hop back 10 → 170. Hop back 3 → 167.**



4. **Where did Hoppy land? 167. That is the answer.**

   181 − 14 = 167 I subtracted 14 in all (1 + 10 + 3).

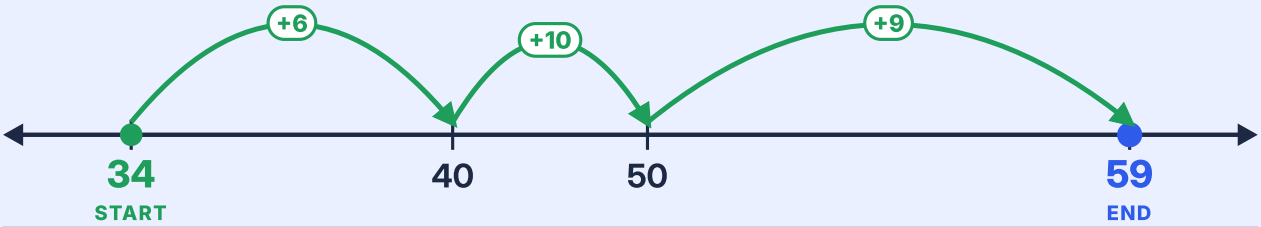
REMEMBER

Kind 2: the answer is **where you land**. Check that the hops add up to the number you took away.

Adding and subtracting: step by step

The **first** number is where you START. The **second** number is **how much to hop in all**. Keep hopping until nothing is left.

ADD: 34 + 25 · START ON THE LEFT, HOP RIGHT, ANSWER ON THE RIGHT

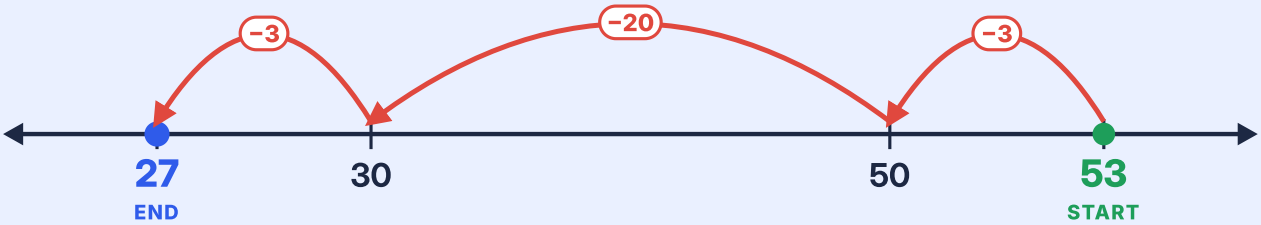


Hop	From → to	Hop size	Still to hop (of 25)
1. to the next ten	34 → 40	6	25 − 6 = 19
2. the tens	40 → 50	10	19 − 10 = 9
3. the rest	50 → 59	9	9 − 9 = 0 ✔ stop

34 + 25 = 59

Check: 6 + 10 + 9 = 25

SUBTRACT: 53 − 26 · START ON THE RIGHT, HOP LEFT, ANSWER ON THE LEFT



Hop	From → to	Hop size	Still to hop (of 26)
1. back to the ten	53 → 50	3	26 − 3 = 23
2. the tens	50 → 30	20	23 − 20 = 3
3. the rest	30 → 27	3	3 − 3 = 0 ✔ stop

53 − 26 = 27

Check: 3 + 20 + 3 = 26

REMEMBER

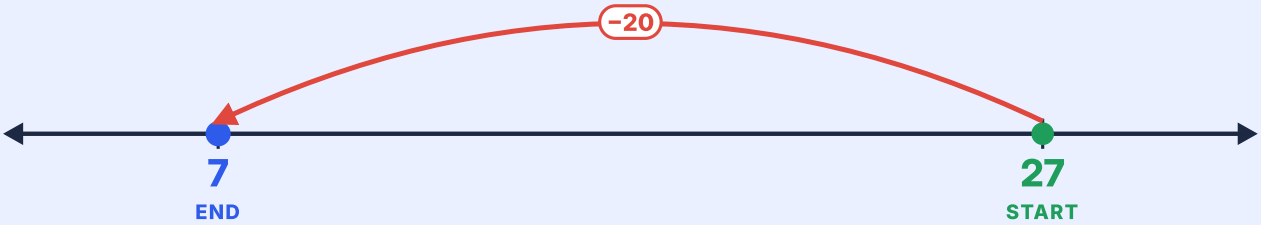
Plus +: START on the left → hop right → the answer is on the right (bigger).

Minus −: START on the right → hop left → the answer is on the left (smaller).

27 – 20 and 27 – 13

27 – 20: ONE HOP OF 20 OR TWO HOPS OF 10?

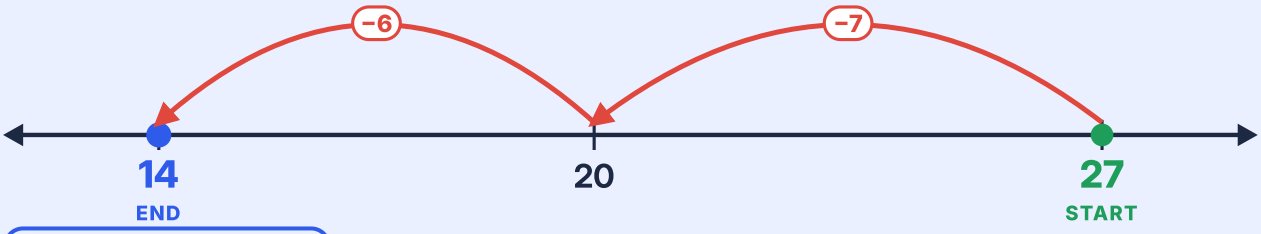
Both are correct. The book often makes **one hop** by tens (like "hop back 20 more" from 50 to 30).



One hop of 20 = two hops of 10. Hoppy lands on **7** both ways.

27 – 13 THE BENCHMARK WAY

27 → 20 needs **7**. 13 = 7 + 6. Hop back 7 → 20, then 6 more → 14.

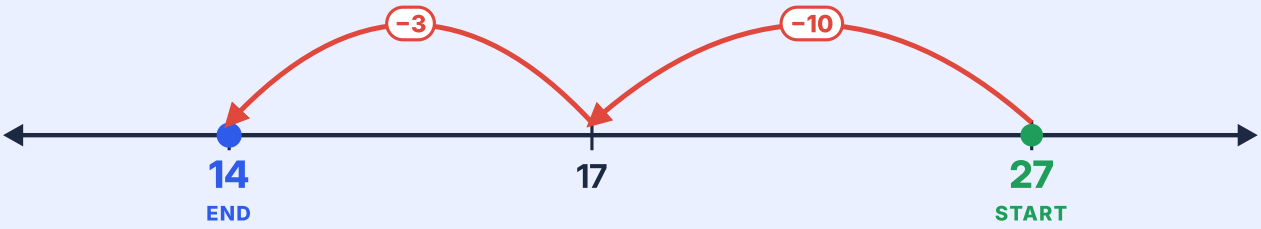


27 – 13 = 14

Check: 7 + 6 = 13 ✔

ANOTHER GOOD WAY: TENS, THEN ONES

27 − 10 → 17, then − 3 → 14. Same answer. Use the benchmark way when your book asks for benchmark numbers.



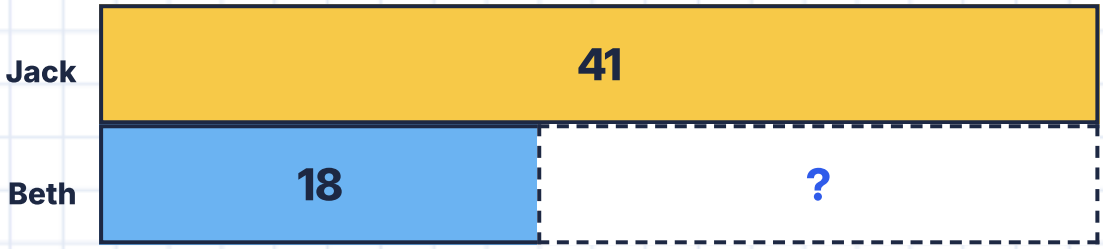
DO IT WITH ME

Let us solve the book's practice together

Jack rides 41 miles in a car. Beth rides 18 miles on a bike. How many more miles does Jack ride than Beth?

1. **Which kind? "How many more" → Kind 1. Add up the hops.**

2. **DRAW the tapes.**



3. **Start at 41. Hop back to a benchmark: 41 → _____. Hop: _____**

4. **Hop back by tens to the ten next to 18: _____ → _____. Hop: _____**

5. **Hop back to 18: _____ → 18. Hop: _____**

Draw here

6. **Add the hops: _____ + _____ + _____ = _____**

Equation: 41 − _____ = 18 Jack rides _____ more miles than Beth.

REMEMBER

The recipe

1. **READ: Is it Kind 1 (how many more / difference) or Kind 2 (take away)?**
2. **DRAW a tape diagram. For Kind 1, two tapes; the difference is the dashed ? part.**
3. **Draw a number line and START at your number.**
   Kind 1: the bigger number (hop back) or the smaller number (count on). Kind 2: the first number.
4. **Hop to a benchmark (a ten) first.**
5. **Hop by tens, then hop the last ones.**
6. **Kind 1: ADD the hops. Kind 2: the answer is where you land.**

COMMON MISTAKES

- Writing the landing number as the answer in a "how many more" question (26 instead of 27).
- Forgetting to add one of the hops.
- Hopping past the benchmark (53 → 40 instead of 53 → 50).
- For Kind 2, not checking that the hops add up to the number taken away.

PRACTICE

Your turn! (1) · How many more?

Draw the tapes and a number line. Hop to a benchmark first. Add the hops.

PROBLEM 1

A red ribbon is 62 cm. A blue ribbon is 35 cm. How much longer is the red ribbon?

Draw here

Hops: _____ + _____ + _____ = _____ Equation: _____

PROBLEM 2

Asha reads 70 pages. Omar reads 44 pages. How many more pages does Asha read?

Draw here

Hops: _____ + _____ + _____ = _____ Equation: _____

PROBLEM 3

A frog jumps 48 cm. A rabbit jumps 75 cm. How much farther does the rabbit jump? (Count on from 48!)

Draw here

Hops: _____ + _____ + _____ = _____ Equation: _____

Your turn! (2) · Take away

Start at the first number. Hop to a benchmark first. The answer is where you land.

PROBLEM 4: 93 − 15 = _____



Break it: 15 = _____ + _____ + _____ Land on: _____

PROBLEM 5: 145 − 13 = _____



Break it: 13 = _____ + _____ + _____ Land on: _____

PROBLEM 6: 64 − 18 = _____



Break it: 18 = _____ + _____ + _____ Land on: _____

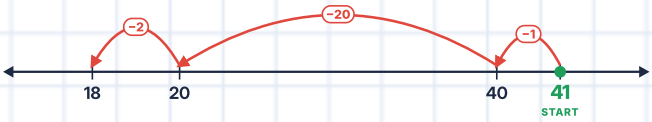
Answers

Step 1: a) 1 · b) 2 · c) 1 **Step 2:** 41: 40, 50 · 18: 10, 20 · 67: 60, 70 · 35: 30, 40

Step 2c: 41 on the right, 18 on the left, benchmarks 40 and 20.

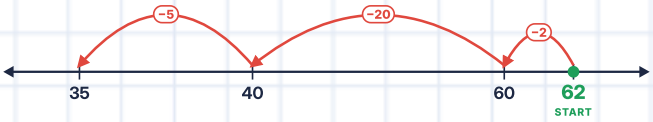
Jack and Beth: 41 → 40 (1), 40 → 20 (20), 20 → 18 (2). 1 + 20 + 2 = **23**. 41 − 23 = 18. Jack rides 23 more miles.

JACK AND BETH: 23



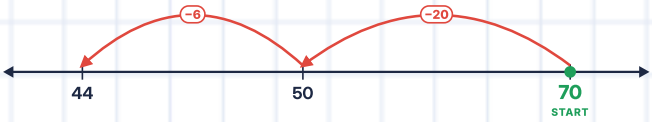
1 + 20 + 2 = 23

1. 62 AND 35: 27



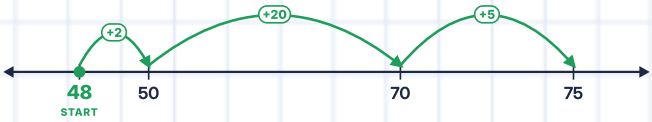
2 + 20 + 5 = 27

2. 70 AND 44: 26



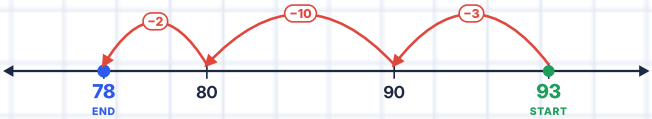
20 + 6 = 26

3. 48 TO 75: 27



2 + 20 + 5 = 27

4. 93 − 15 = 78



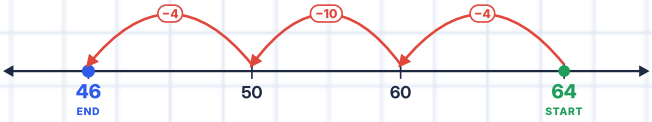
3 + 10 + 2 = 15

5. 145 − 13 = 132



5 + 8 = 13

6. 64 − 18 = 46



4 + 10 + 4 = 18